

Qualification Pack



Hand Soldering Technician - Telecom Boards

QP Code: TEL/Q2500

Version: 4.0

NSQF Level: 3

Telecom || 3rd Floor, Plot No 126, Sector - 44
Gurgaon - 122003 || email:standards@tsscindia.com

Qualification Pack

Contents

TEL/Q2500: Hand Soldering Technician - Telecom Boards	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
TEL/N2500: High-Density Hand Soldering of Components on Telecom Boards	5
TEL/N2501: Rework on Defects and Undertake Selective Soldering	10
TEL/N2502: Cleaning and Inspection of PCBs	15
TEL/N9107: Follow sustainability practices in telecom production and assembly line processes	20
DGT/VSQ/N0101: Employability Skills (30 Hours)	25
Assessment Guidelines and Weightage	30
<i>Assessment Guidelines</i>	30
<i>Assessment Weightage</i>	31
Acronyms	32
Glossary	34

Qualification Pack

TEL/Q2500: Hand Soldering Technician - Telecom Boards

Brief Job Description

A Hand Soldering Technician -Telecom Boards is responsible for soldering and desoldering components, reworking defects, and cleaning and inspecting telecom boards while following the applicable quality and safety standards.

Personal Attributes

The role requires precision, attention to detail, and steady hand-eye coordination. The individual should be process-oriented, focused, and capable of working with concentration for long hours.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N2500: High-Density Hand Soldering of Components on Telecom Boards](#)
2. [TEL/N2501: Rework on Defects and Undertake Selective Soldering](#)
3. [TEL/N2502: Cleaning and Inspection of PCBs](#)
4. [TEL/N9107: Follow sustainability practices in telecom production and assembly line processes](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Handset
Occupation	Communication Electronics
Country	India
NSQF Level	3
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3114.1404

Qualification Pack

Minimum Educational Qualification & Experience	10th grade pass OR 8th grade pass with 3 Years of experience in communication electronics OR Previous relevant Qualification of NSQF Level 2.5 with 1.5 years of experience in communication electronics
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	4.0
Reference code on NQR	QG-03-TL-04092-2025-V2-TSSC
NQR Version	2

Qualification Pack

TEL/N2500: High-Density Hand Soldering of Components on Telecom Boards

Description

This unit is about performing high-density hand soldering of components on telecom boards while ensuring quality, safety, and efficiency.

Scope

The scope covers the following :

- Prepare board and material/components
- Solder components on boards

Elements and Performance Criteria

Prepare board and material/components

To be competent, the user/individual on the job must be able to:

- PC1.** interpret Computer-Aided Design (CAD) specifications to identify soldering requirements
- PC2.** set up the soldering jig/fixture according to work specifications
- PC3.** collect all required components, ensuring their compatibility with the board layout
- PC4.** adjust the soldering station temperature as per the component and board specifications and verify accuracy using sensors
- PC5.** select the appropriate solder bit, soldering wire, and flux based on work specifications
- PC6.** clean soldering bits, component leads, and PCB surfaces to remove oxidation, dust, or contamination
- PC7.** check and clean the solder wire to prevent impurities affecting the soldering process
- PC8.** verify and apply the correct flux to improve soldering quality and prevent defects

Solder components on boards

To be competent, the user/individual on the job must be able to:

- PC9.** position and align components precisely according to specifications, ensuring correct orientation
- PC10.** apply the correct amount of solder, flux, and heat, ensuring even flow without excess solder on gold-plated areas
- PC11.** control soldering temperature and nozzle vector speed to maintain uniformity and avoid thermal damage to components
- PC12.** follow proper soldering techniques to minimize solder bridges, cold joints, and dry soldering defects
- PC13.** maintain the soldering station by performing start-up, shutdown, and preventive maintenance procedures
- PC14.** check solder joints visually or using magnification tools to ensure compliance with industry standards

Qualification Pack

- PC15.** rectify minor soldering defects, such as excess solder, bridges, or cold joints, using appropriate rework techniques

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Standard Operating Procedures (SOPs) for high-density soldering in telecom applications
- KU2.** the risks of improper soldering, including cold joints, short circuits, and component damage
- KU3.** industry standards such as IPC-A-610 for soldering quality and acceptability criteria
- KU4.** methods for preventing contamination of PCBs, components, and soldering materials
- KU5.** the procedures for cleaning and maintaining soldering tools, fixtures, and workstations
- KU6.** the importance of controlling temperature, humidity, and electrostatic discharge (ESD) in the soldering process
- KU7.** techniques for handling delicate components and avoiding thermal or mechanical stress
- KU8.** different types of soldering defects, their causes, and corrective actions
- KU9.** documentation and record-keeping requirements related to soldering and board assembly
- KU10.** Safety, Health, and Environmental (SHE) guidelines, including the safe handling of lead-based and lead-free soldering materials

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret CAD layouts, work instructions, and technical manuals accurately
- GS2.** document work completed, maintain soldering logs, and report any quality concerns
- GS3.** communicate effectively with supervisors, team members, and quality inspectors
- GS4.** identify and troubleshoot minor soldering defects before the final inspection
- GS5.** follow safety procedures, including ESD handling and hazardous material disposal
- GS6.** plan work sequences efficiently to minimize errors and improve productivity
- GS7.** use basic measuring tools, such as calipers and temperature sensors, to verify precision
- GS8.** adapt to different soldering materials and techniques as per evolving industry standards
- GS9.** maintain concentration and hand-eye coordination for precision soldering
- GS10.** collaborate with colleagues to improve workflow and ensure quality compliance

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare board and material/components</i>	16	20	-	10
PC1. interpret Computer-Aided Design (CAD) specifications to identify soldering requirements	2	2	-	1
PC2. set up the soldering jig/fixture according to work specifications	2	2	-	1
PC3. collect all required components, ensuring their compatibility with the board layout	2	3	-	2
PC4. adjust the soldering station temperature as per the component and board specifications and verify accuracy using sensors	2	3	-	2
PC5. select the appropriate solder bit, soldering wire, and flux based on work specifications	2	3	-	1
PC6. clean soldering bits, component leads, and PCB surfaces to remove oxidation, dust, or contamination	2	3	-	1
PC7. check and clean the solder wire to prevent impurities affecting the soldering process	2	2	-	1
PC8. verify and apply the correct flux to improve soldering quality and prevent defects	2	2	-	1
<i>Solder components on boards</i>	14	30	-	10
PC9. position and align components precisely according to specifications, ensuring correct orientation	2	4	-	1
PC10. apply the correct amount of solder, flux, and heat, ensuring even flow without excess solder on gold-plated areas	2	4	-	1
PC11. control soldering temperature and nozzle vector speed to maintain uniformity and avoid thermal damage to components	2	5	-	2
PC12. follow proper soldering techniques to minimize solder bridges, cold joints, and dry soldering defects	2	5	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. maintain the soldering station by performing start-up, shutdown, and preventive maintenance procedures	2	4	-	2
PC14. check solder joints visually or using magnification tools to ensure compliance with industry standards	2	4	-	1
PC15. rectify minor soldering defects, such as excess solder, bridges, or cold joints, using appropriate rework techniques	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N2500
NOS Name	High-Density Hand Soldering of Components on Telecom Boards
Sector	Telecom
Sub-Sector	Handset
Occupation	Communication Electronics
NSQF Level	3
Credits	5
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N2501: Rework on Defects and Undertake Selective Soldering

Description

This unit covers the process of reworking defects and performing selective soldering of components on telecom boards, ensuring compliance with specifications and quality standards.

Scope

The scope covers the following :

- Verification of completed board against specifications
- Selective soldering and rework

Elements and Performance Criteria

Verification of completed board against specifications

To be competent, the user/individual on the job must be able to:

- PC1.** retrieve and interpret the correct CAD layout and BOM from the software library/system
- PC2.** compare and verify on-board component placement and orientation with CAD specifications
- PC3.** cross-check component values and specifications against drawings/work orders
- PC4.** check the board for defects such as excess burr on component bases, visible damage, solder bridges, pinholes/blowholes, spikes, peaks, gold fingers, and dry solder joints
- PC5.** conduct and interpret functional tests of the board using appropriate tools and equipment

Selective soldering and rework

To be competent, the user/individual on the job must be able to:

- PC6.** identify and mark defective components for rework or replacement
- PC7.** determine whether components can be reused or require sourcing new replacements
- PC8.** remove faulty solder joints using appropriate desoldering techniques and tools
- PC9.** apply proper selective soldering techniques, maintaining appropriate temperature and solder flow
- PC10.** ensure compliance with soldering specifications, avoiding overheating, cold solder joints, and improper wetting
- PC11.** reassemble and resolder components while adhering to ESD safety precautions
- PC12.** verify reworked components for correct placement, orientation, and connectivity under a microscope
- PC13.** maintain records of reworked components and report unresolved defects per escalation protocols

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** industry standards and best practices for telecom board soldering and rework

Qualification Pack

- KU2.** procedures for accessing and interpreting CAD drawings and BOMs
- KU3.** quality control checks and defect identification techniques in PCB assembly
- KU4.** use and maintenance of soldering stations, desoldering tools, and rework stations
- KU5.** work process documentation and escalation matrix for defect reporting
- KU6.** ESD control measures and their importance in preventing component damage
- KU7.** Occupational health and safety (OHS) and environmental regulations applicable to soldering operations
- KU8.** methods for assessing solder joint quality, including visual inspection, X-ray, and automated optical inspection (AOI)
- KU9.** compatibility of solder types, flux, and component materials in rework and selective soldering
- KU10.** industry protocols for handling lead-free and lead-based soldering materials

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret CAD drawings, BOMs, and work instructions accurately
- GS2.** follow standard operating procedures (SOPs) and safety guidelines effectively
- GS3.** communicate clearly with team members, supervisors, and quality inspectors
- GS4.** maintain detailed documentation of rework activities and defect reports
- GS5.** use digital tools for defect tracking, component sourcing, and reporting
- GS6.** identify and troubleshoot minor soldering defects independently
- GS7.** handle and store electronic components safely, following ESD precautions
- GS8.** adapt to new soldering technologies and industry advancements
- GS9.** collaborate with other departments to resolve quality-related issues
- GS10.** manage time efficiently to meet production deadlines and quality standards

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Verification of completed board against specifications</i>	16	20	-	10
PC1. retrieve and interpret the correct CAD layout and BOM from the software library/system	3	4	-	2
PC2. compare and verify on-board component placement and orientation with CAD specifications	3	4	-	2
PC3. cross-check component values and specifications against drawings/work orders	4	4	-	2
PC4. check the board for defects such as excess burr on component bases, visible damage, solder bridges, pinholes/blowholes, spikes, peaks, gold fingers, and dry solder joints	4	4	-	2
PC5. conduct and interpret functional tests of the board using appropriate tools and equipment	2	4	-	2
<i>Selective soldering and rework</i>	14	30	-	10
PC6. identify and mark defective components for rework or replacement	1	3	-	1
PC7. determine whether components can be reused or require sourcing new replacements	2	4	-	1
PC8. remove faulty solder joints using appropriate desoldering techniques and tools	2	4	-	2
PC9. apply proper selective soldering techniques, maintaining appropriate temperature and solder flow	2	4	-	2
PC10. ensure compliance with soldering specifications, avoiding overheating, cold solder joints, and improper wetting	2	4	-	1
PC11. reassemble and resolder components while adhering to ESD safety precautions	2	4	-	1
PC12. verify reworked components for correct placement, orientation, and connectivity under a microscope	2	4	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. maintain records of reworked components and report unresolved defects per escalation protocols	1	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N2501
NOS Name	Rework on Defects and Undertake Selective Soldering
Sector	Telecom
Sub-Sector	Handset
Occupation	Communication Electronics
NSQF Level	3
Credits	5
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N2502: Cleaning and Inspection of PCBs

Description

This unit covers the process of cleaning and inspecting completed PCBs to ensure quality, reliability, and compliance with industry standards

Scope

The scope covers the following :

- Cleaning of PCBs
- Quality checks and assurance

Elements and Performance Criteria

Cleaning of PCBs

To be competent, the user/individual on the job must be able to:

- PC1.** clean PCBs to remove flux residues, white patches, and other contaminants using specified solvents and cleaning agents
- PC2.** operate vapor de-greaser, ultrasonic cleaner, and other industry-recommended equipment for cleaning PCBs, ensuring effective removal of impurities
- PC3.** dry PCBs using controlled methods to prevent damage and contamination
- PC4.** package and store cleaned PCBs using electrostatic discharge (ESD)-safe materials and procedures to maintain integrity and prevent damage

Quality checks and assurance

To be competent, the user/individual on the job must be able to:

- PC5.** check PCBs for missing, wrongly mounted, or misaligned components as per specifications and bill of materials (BOM)
- PC6.** inspect soldering quality, including workmanship, joint integrity, and adherence to IPC standards
- PC7.** verify placement of board identifiers, conformal coatings, and compliance with customer or regulatory requirements
- PC8.** identify defects such as cold solder joints, bridging, and insufficient solder, and take corrective action
- PC9.** identify and segregate functional and defective PCBs
- PC10.** document findings and escalate non-conformities as per established reporting protocols
- PC11.** carry out routine maintenance of the SMT equipment to ensure their optimal functioning
- PC12.** conduct routine shop floor audits to ensure adherence to process parameters, including temperature and humidity control

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** work/project management principles and their application in PCB inspection
- KU2.** risks and consequences of non-compliance with defined procedures
- KU3.** escalation matrix for reporting incidents such as system failures, contamination issues, and non-conformities
- KU4.** importance of record-keeping and potential consequences of poor documentation
- KU5.** Safety, Health, and Environment (SHE) and Occupational Health and Safety (OHS) regulations as per company and industry standards
- KU6.** properties, usage, and handling of cleaning chemicals/solvents used for PCB cleaning
- KU7.** principles and techniques of PCB cleaning, including solvent-based and ultrasonic methods
- KU8.** operation and maintenance of cleaning equipment such as vapor de-greasers and ultrasonic cleaners
- KU9.** quality control and quality assurance parameters related to PCB inspection
- KU10.** IPC and industry standards for soldering, component placement, and PCB integrity
- KU11.** ESD-safe handling procedures and their impact on PCB reliability

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret technical documents, including schematics, BOMs, and work instructions
- GS2.** comprehend and apply safety manuals, operational guidelines, and quality standards
- GS3.** communicate effectively with team members, supervisors, and external stakeholders
- GS4.** problem-solving skills to identify, analyze, and resolve quality or process-related issues
- GS5.** practice attention to detail to detect defects, inconsistencies, and non-conformities in PCBs
- GS6.** collaborate effectively in a production and quality control environment
- GS7.** adapt to learn and implement new inspection techniques, tools, and industry best practices
- GS8.** manage time effectively to meet production deadlines and quality standards
- GS9.** use digital tools and software for data recording, reporting, and analysis
- GS10.** ensure quality standards align with end-user requirements and expectations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cleaning of PCBs</i>	10	20	-	6
PC1. clean PCBs to remove flux residues, white patches, and other contaminants using specified solvents and cleaning agents	2	5	-	2
PC2. operate vapor de-greaser, ultrasonic cleaner, and other industry-recommended equipment for cleaning PCBs, ensuring effective removal of impurities	3	5	-	2
PC3. dry PCBs using controlled methods to prevent damage and contamination	3	5	-	1
PC4. package and store cleaned PCBs using electrostatic discharge (ESD)-safe materials and procedures to maintain integrity and prevent damage	2	5	-	1
<i>Quality checks and assurance</i>	20	30	-	14
PC5. check PCBs for missing, wrongly mounted, or misaligned components as per specifications and bill of materials (BOM)	3	4	-	2
PC6. inspect soldering quality, including workmanship, joint integrity, and adherence to IPC standards	3	4	-	2
PC7. verify placement of board identifiers, conformal coatings, and compliance with customer or regulatory requirements	3	4	-	2
PC8. identify defects such as cold solder joints, bridging, and insufficient solder, and take corrective action	3	4	-	2
PC9. identify and segregate functional and defective PCBs	2	4	-	2
PC10. document findings and escalate non-conformities as per established reporting protocols	2	3	-	1
PC11. carry out routine maintenance of the SMT equipment to ensure their optimal functioning	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. conduct routine shop floor audits to ensure adherence to process parameters, including temperature and humidity control	2	4	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N2502
NOS Name	Cleaning and Inspection of PCBs
Sector	Telecom
Sub-Sector	Handset
Occupation	Communication Electronics
NSQF Level	4
Credits	3
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N9107: Follow sustainability practices in telecom production and assembly line processes

Description

This unit covers sustainability practices in telecom production and assembly lines. It includes identification, segregation, recycling, and responsible disposal of telecom components and materials, ensuring compliance with environmental regulations and promoting sustainable manufacturing processes.

Scope

The scope covers the following :

- Identify and segregate recyclable, refurbishable, and hazardous components
- Follow green manufacturing and sustainable assembly practices
- Dispose of and recycle waste
- Comply with environmental and sustainability regulations

Elements and Performance Criteria

Identify and segregate recyclable, refurbishable, and hazardous components

To be competent, the user/individual on the job must be able to:

- PC1.** inspect incoming and outgoing materials to identify recyclable, refurbishable, and hazardous components
- PC2.** categorize telecom components such as PCBs, cables, batteries, and plastic casings for appropriate processing
- PC3.** maintain an inventory of recyclable and hazardous materials to track waste management
- PC4.** store e-waste separately and label it in accordance with Extended Producer Responsibility (EPR) guidelines
- PC5.** identify and report any improper disposal practices to the designated authority

Follow green manufacturing and sustainable assembly practices

To be competent, the user/individual on the job must be able to:

- PC6.** use energy-efficient equipment, tools, and automated systems to reduce carbon footprint
- PC7.** follow low-emission soldering and lead-free assembly processes to minimize pollution
- PC8.** optimize the use of materials, water, and electricity during production
- PC9.** conduct regular maintenance and calibration of energy-efficient machinery to ensure optimal performance
- PC10.** adopt ISO 14001 (Environmental Management System) compliance in production workflows

Dispose of and recycle waste

To be competent, the user/individual on the job must be able to:

- PC11.** dispose of hazardous waste (e.g., lithium batteries, chemical residues) in accordance with environmental safety protocols
- PC12.** deposit non-hazardous, recyclable materials (e.g., plastics, aluminum, copper) in designated collection areas

Qualification Pack

- PC13.** maintain documentation for waste disposal and recycling compliance audits
- PC14.** conduct periodic waste audits to identify opportunities for further waste reduction
- PC15.** coordinate with authorized e-waste recyclers to ensure proper processing and disposal

Comply with environmental and sustainability regulations

To be competent, the user/individual on the job must be able to:

- PC16.** follow the applicable national and international environmental laws and regulations for telecom manufacturing
- PC17.** participate in sustainability audits and assessments as per company policies
- PC18.** check for adherence to sustainability guidelines and report deviations
- PC19.** follow corrective actions based on environmental audits and assessments

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization's sustainability policies and goals
- KU2.** EPR guidelines, ISO 14001, and e-waste disposal laws
- KU3.** the methods for identifying recyclable and hazardous components
- KU4.** green manufacturing practices, including energy-efficient tools, lead-free soldering, and automation
- KU5.** proper handling, storage, and disposal of e-waste
- KU6.** the techniques for reducing material wastage without impacting production quality
- KU7.** safe and responsible handling of hazardous and non-hazardous materials
- KU8.** environmental impact assessment techniques for telecom production
- KU9.** the documentation requirements for sustainability audits and compliance tracking

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret technical manuals, regulatory guidelines, and environmental compliance documents
- GS2.** effectively report waste management data, sustainability initiatives, and compliance issues
- GS3.** identify and implement solutions to reduce material wastage and improve resource efficiency
- GS4.** coordinate with waste management teams, auditors, and suppliers to implement sustainable practices
- GS5.** assess and interpret sustainability data to improve environmental impact measures
- GS6.** evaluate options for waste management and select the most sustainable alternative

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify and segregate recyclable, refurbishable, and hazardous components</i>	6	10	-	5
PC1. inspect incoming and outgoing materials to identify recyclable, refurbishable, and hazardous components	1	2	-	1
PC2. categorize telecom components such as PCBs, cables, batteries, and plastic casings for appropriate processing	1	2	-	1
PC3. maintain an inventory of recyclable and hazardous materials to track waste management	1	2	-	1
PC4. store e-waste separately and label it in accordance with Extended Producer Responsibility (EPR) guidelines	1	2	-	1
PC5. identify and report any improper disposal practices to the designated authority	2	2	-	1
<i>Follow green manufacturing and sustainable assembly practices</i>	10	15	-	5
PC6. use energy-efficient equipment, tools, and automated systems to reduce carbon footprint	2	3	-	1
PC7. follow low-emission soldering and lead-free assembly processes to minimize pollution	2	3	-	1
PC8. optimize the use of materials, water, and electricity during production	2	3	-	1
PC9. conduct regular maintenance and calibration of energy-efficient machinery to ensure optimal performance	2	3	-	1
PC10. adopt ISO 14001 (Environmental Management System) compliance in production workflows	2	3	-	1
<i>Dispose of and recycle waste</i>	10	15	-	6
PC11. dispose of hazardous waste (e.g., lithium batteries, chemical residues) in accordance with environmental safety protocols	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. deposit non-hazardous, recyclable materials (e.g., plastics, aluminum, copper) in designated collection areas	2	3	-	2
PC13. maintain documentation for waste disposal and recycling compliance audits	2	3	-	1
PC14. conduct periodic waste audits to identify opportunities for further waste reduction	2	3	-	1
PC15. coordinate with authorized e-waste recyclers to ensure proper processing and disposal	2	3	-	1
<i>Comply with environmental and sustainability regulations</i>	4	10	-	4
PC16. follow the applicable national and international environmental laws and regulations for telecom manufacturing	1	2	-	1
PC17. participate in sustainability audits and assessments as per company policies	1	3	-	1
PC18. check for adherence to sustainability guidelines and report deviations	1	3	-	1
PC19. follow corrective actions based on environmental audits and assessments	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N9107
NOS Name	Follow sustainability practices in telecom production and assembly line processes
Sector	Telecom
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	22/01/2026
Next Review Date	22/01/2029
NSQC Clearance Date	22/01/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 50

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N2500.High-Density Hand Soldering of Components on Telecom Boards	30	50	-	20	100	25
TEL/N2501.Rework on Defects and Undertake Selective Soldering	30	50	-	20	100	25
TEL/N2502.Cleaning and Inspection of PCBs	30	50	0	20	100	20
TEL/N9107.Follow sustainability practices in telecom production and assembly line processes	30	50	0	20	100	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	140	230	0	80	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
TSSC	Telecom Sector Skill Council
TRAI	Telecom Regulatory Authority of India
PCB	Printed Circuit Board
ESD	Electrostatic Discharge
SMT	Surface-Mount Technology
THT	Through-Hole Technology
RoHS	Restriction of Hazardous Substances
IPC	Institute for Printed Circuits (Now IPC -Association Connecting Electronics Industries)
BGA	Ball Grid Array
AOI	Automated Optical Inspection
X-Ray	X-Ray Inspection (for PCB quality checks)
ICT	In-Circuit Testing
MSD	Moisture-Sensitive Device

Qualification Pack

REACH

Registration, Evaluation, Authorisation, and Restriction of Chemicals

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.

Qualification Pack

National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.